

# **Encryption and Decryption Tool for Securing Data**

## **1. Problem Statement**

In today's digital world, data security has become a major concern. Sensitive information such as personal files, passwords, and confidential documents are at risk of unauthorized access. With the increasing number of cyber attacks and data breaches, protecting data has become essential.

There is a need for a system that can secure files and prevent unauthorized users from accessing or reading sensitive data.

## **2. Project Objective**

The main objective of this project is to develop a tool that:

- Encrypts files to make them secure
- Decrypts files to restore them to their original form
- Provides a user-friendly interface
- Protects data from unauthorized access

## **3. Solution Description**

In this project, a GUI-based application is developed using Python. The application uses the cryptography library to perform encryption and decryption of files.

### **Working Process:**

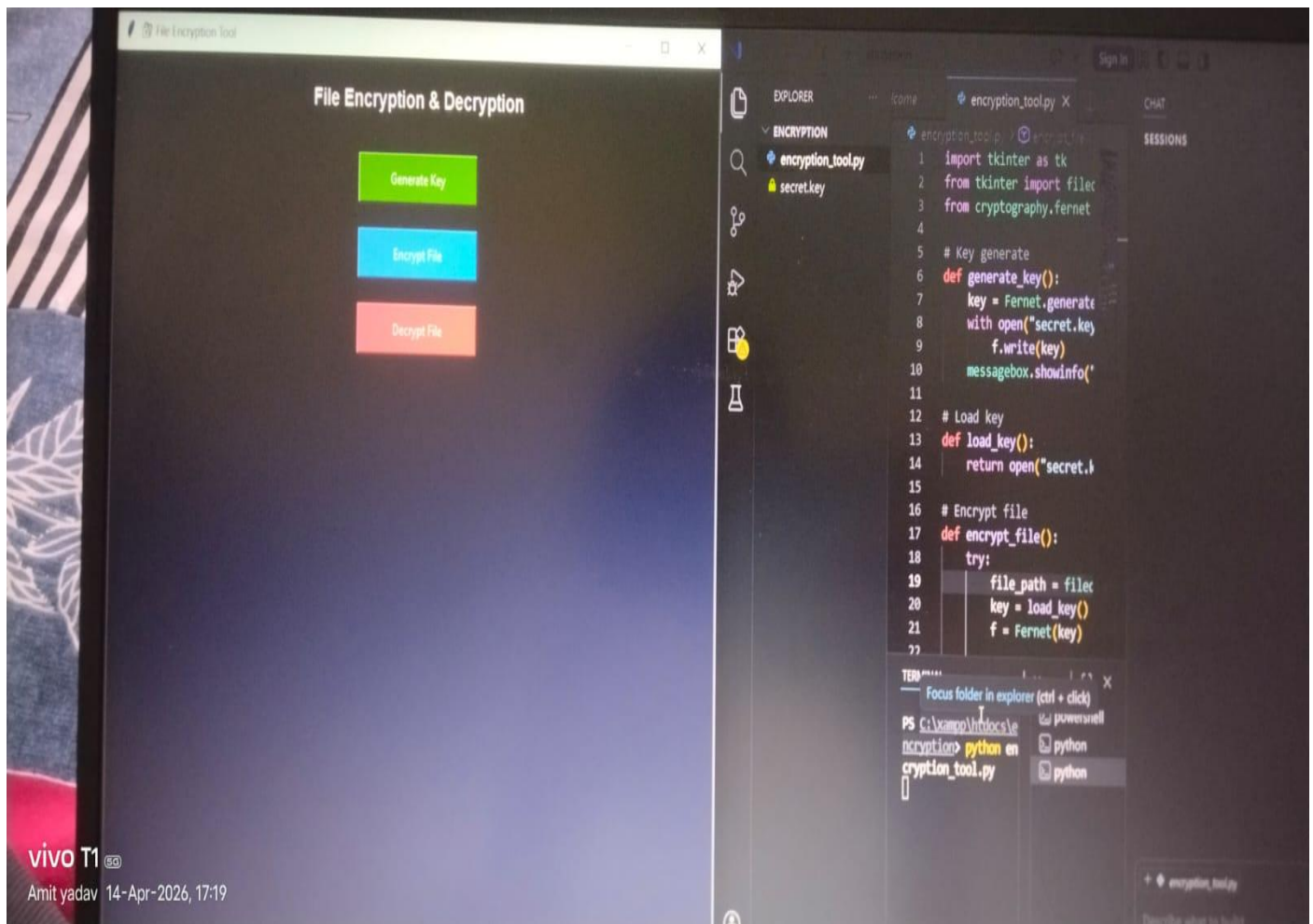
- A secret key is generated
- The user selects a file
- The encryption process converts the file into an unreadable format
- The decryption process uses the same key to restore the file to its original format

### **The graphical user interface (GUI) includes buttons such as:**

- Generate Key
- Encrypt File
- Decrypt File

This system is simple, efficient, and ensures data security.

## 4. Application Screenshots



## 5. Open Source Code Link

The project has been successfully uploaded to GitHub.

GitHub Repository Link:

<https://github.com/Munitayadav/file-encryption-tool>

## 6. Future Scope

This project can be further improved in the future by:

- Adding password-based encryption
- Integrating cloud storage
- Developing a mobile application version
- Supporting multiple file encryption
- Implementing advanced security algorithms

## Conclusion

In this project, we developed a simple and effective File Encryption and Decryption Tool that helps in securing data. It prevents unauthorized access and demonstrates the basic concepts of cyber security in a practical way.